


Islamic Emirates of Afghanistan

Ministry of Higher Education

Kabul university

Computer Science Faculty

Software Engineering Department

Teacher: Khwajah Hameedullah

Subject: Programing Fundamentals

Topic: Chapter 2 checkpoints and unit quiz

Ali Reza "Rezwani"

**2.3.1** How do you write a statement to let the user enter a double value from the keyboard? What happens if you entered **5a** when executing the following code?

```
double radius = input.nextDouble();
```

We can use Scanner statement to allow users enter any information, so for entering a double value which includes decimal values we can use following code:

```
import java.util.Scanner;
public class basic {
    public static void main(String[] args){
        Scanner scanner = new Scanner(System.in);
        double x = scanner.nextDouble();
```

**Answer**: if user write 5a in a double type the following error will happen: which is a runtime error Exception in thread "main" java.util.InputMismatchException.

**2.3.2** Are there any performance differences between the following two **import** statements?

```
import java.util.Scanner;
import java.util.*;
```

The first one is an especial kind of the second one. If we use the second one it contains all classes but when we use Scanner class it is just for scanner class not for other classes.

**2.4.1** Which of the following identifiers are valid? Which are Java keywords?

```
miles, Test, a++, --a, 4#R, $4, #44, apps
class, public, int, x, y, radius
```

**Answer**:

a. These are valid: miles, Test, $4, apps, x, y, radius
b. These are not valid: a++, --a, 4#R, #44, class, public, int

**2.5.1** Identify and fix the errors in the following code:

```
1  public class Test {
2    public static void main(String[] args) {
3      int i = k + 2;
4      System.out.println(i);
5    }
6  }
```

**Answer:** In the above code data type is defined as an integer but written value id a string not a whole number. So it can be written like:

```
public class Test {
    public static void main(String[] args) {
        int i = 2;
        int k = 3;
        System.out.print(i + k);
```

**2.6.1** Identify and fix the errors in the following code:

```
1  public class Test {
2    public static void main(String[] args) {
3      int i = j = k = 2;
4      System.out.println(i + " " + j + " " + k);
5    }
6  }
```

**Answer:** Here String values typed in an integer type variable so it should be declared that (j and k) are integers. The code can be written like:

```
public class Test {
    public static void main(String[] args) {
        int i , k ,j;
        i = k = j = 2;
        System.out.print(i +" " + k + " "  + j);
    }
}
```

**2.7.1** What are the benefits of using constants? Declare an `int` constant `SIZE` with value `20`.

**Answer:** Using constants are beneficial when we are working with mathematical constants like: Euler number, Pi, etc.

```
public class Test {
    public static void main(String[] args) {
        final int SIZE = 20;
        System.out.print(SIZE);
    }
}
```

**2.7.2** Translate the following algorithm into Java code:

Step 1: Declare a `double` variable named `miles` with an initial value `100`.

Step 2: Declare a `double` constant named `KILOMETERS_PER_MILE` with value `1.609`.

Step 3: Declare a `double` variable named `kilometers`, multiply `miles` and `KILOMETERS_PER_MILE`, and assign the result to `kilometers`.

Step 4: Display `kilometers` to the console.

What is `kilometers` after Step 4?

**Answer:** The kilometers is the output here.

```
public class Test {
    public static void main(String[] args) {
        double miles = 100;
        final double KILOMETERS_PER_MILES = 1.609;
        double Kilometers = miles * KILOMETERS_PER_MILES;
        System.out.print(miles + " miles is equal to "+Kilometers + " kilometers");
    }
}
```

**2.8.1** What are the naming conventions for class names, method names, constants, and variables? Which of the following items can be a constant, a method, a variable, or a class according to the Java naming conventions?

`MAX_VALUE`, `Test`, `read`, `readDouble`

**Answer:** MAX_VALUE, Test, read, readDouble
MAX_VALUE…..can be constant name readDouble…….can be variables name
Test……..…can be class name
Read……..…can be methods name or variables name

**2.9.1** Find the largest and smallest **byte**, **short**, **int**, **long**, **float**, and **double**. Which of these data types requires the least amount of memory?

**Answer:**
Byte……..(-128-127), short(-32768 to 32767), int(-2147483648 to 2147483647)
Long(-9223372036854775808 to 9223372036854775807), float(6 to 7) decimal digits, double(15 to 16).
The least amount of memory is for byte data type which stores 1byte(8 bits).

**2.9.2** Show the result of the following remainders:

**Answer:** 56 % 6 , 78 % -4 , -34 % 5 , -34 % -5 , 5 % 1 , 1 % 5

```
public class Test {
    public static void main(String[] args) {
        int a, b, c, d, e, f;
        a =  56 % 6;     b = 78 % -4;      c = -34 % 5;      d = -34 % -5;
        e = 5 % 1;     f = 1 % 5;
        System.out.print(a +" "+ b +" "+ c +" " + d + " " + e + " " + f);
    }
}
```

**2.9.3** If today is Tuesday, what will be the day in 100 days?

**Answer:** Tuesday is the fourth day of the week we can divide 100 to 7 to know what is the remainder.

```
public class Test {
    public static void main(String[] args) {
        System.out.println((100 + 4) % 7);
```

As the result here is equal to 6 so know that Thursday is the sixth day of the week.
Another way to do this task is using the if class to say that as an example if the remainder is 6 so print for as the sixth day of the week which is Thursday.

**2.9.4** What is the result of **25 / 4**? How would you rewrite the expression if you wished the result to be a floating-point number?

**Answer:**

```
public class Test {
    public static void main(String[] args) {
        double result = 25.0 / 4;
        System.out.print(result);
    }
}
```

**2.9.5** Show the result of the following code:

**Answer:**
System.out.println(2 * (5 / 2 + 5 / 2)); 8
System.out.println(2 * (5 / 2)); 4
System.out.println(2 * 5 / 2 + 2 * 5 / 2); 10
System.out.println(2 * 5 / 2); 5

**2.9.6** Are the following statements correct? If so, show the output.

```
System.out.println("25 / 4 is " + 25 / 4);
System.out.println("25 / 4.0 is " + 25 / 4.0);
System.out.println("3 * 2 / 4 is " + 3 * 2 / 4);
System.out.println("3.0 * 2 / 4 is " + 3.0 * 2 / 4);
```

**Answer:**

```
25 / 4 is 6
25 / 4.0 is 6.25
3 * 2 / 4 is 1
3.0 * 2 / 4 is 1.5
```

Yes, the all above statements are true and the output is:

**2.9.7** Write a statement to display the result of $2^{3.5}$.

**Answer:**

```
public class Test {
    public static void main(String[] args) {
        System.out.println(Math.pow(2, 3.5));
    }
}
```

**2.9.8** Suppose m and r are integers. Write a Java expression for $mr^2$ to obtain a floating-point result.

**Answer:**

```
public class Test {
    public static void main(String[] args) {
        int m = 0;          int r = 0;
        System.out.print(m * Math.pow(r, 2));
    }
}
```

**2.10.1** How many accurate digits are stored in a `float` or `double` type variable?

**Answer:**

For float type 6-9 accurate digits can be stored
For double type 15-17 accurate digits can be stored

**2.10.2** Which of the following are correct literals for floating-point numbers?

`12.3, 12.3e+2, 23.4e-2, –334.4, 20.5, 39F, 40D`

**Answer:** All above given numbers are correct literals for floating-point numbers.
12.3 , -334.4 , 20.5 , are standard floating-point numbers; 12.3e+2 , 23.4e-2 , are written in scientific form and 39f in which (f) is for float and 40D in which (D) is for double.

**2.10.3** Which of the following are the same as `52.534`?

`5.2534e+1, 0.52534e+2, 525.34e–1, 5.2534e+0`

**Answer:** In all above given floating-point literals jut the last one (2534e+0) is not equal to 52.534 ; so other given numeric literals are exactly equal to 52.534 .

**2.10.4** Which of the following are correct literals?

`5_2534e+1, _2534, 5_2, 5_`

**Answer:** 5_2534e+ and 5_2 are correct literals, because java allow us to use under score for more readability. But _2534 as well as 5_ are not correct literals, because the under score in java which is for mor readability must be between tow digits not at the end o at the first.

## 2.11.1 What does REPL stand for? How do you launch JShell?

**Answer:** REPL stands for (Read_Evaluate_Print_Loop). We can launch JShell by entering jshell in a command window in our computer.

## 2.12.1 How would you write the following arithmetic expressions in Java?

a. $\frac{4}{3(r + 34)} - 9(a + bc) + \frac{3 + d(2 + a)}{a + bd}$

b. $5.5 \times (r + 2.5)^{2.5+t}$

**Answer:**

a. As we know that Arithmetic expressions is same as java expressions so we can write is in java like following:

```
b. public class CheckPoints {
       public static void main(String[] args){
           double r=0, a=0, b=0, c=0, d=0;
           double result = 4/(3 * (r + 34)) -9 * (a + b * c) +(3 + d * (2+a))/(a +
   b*d);
           System.out.println(result);
       }
   }
```

c. in this expression we have some in power so we can use Math.pow method to write it in java:

```
public class CheckPoints {
    public static void main(String[] args){
        double r=0, t=0;
        System.out.println(5.5 * Math.pow((r+2.5), (2.5 + t)));
    }
}
```

## 2.13.1 How do you obtain the current second, minute, and hour?

**Answer:** current time can be obtained as following:

```
public class CheckPoints {
    public static void main(String[] args){
        long currentMilliSeconds = System.currentTimeMillis();
        long totalSeconds = currentMilliSeconds / 1000;
        long currentSeconds = totalSeconds % 60;
        long totalMinutes = totalSeconds / 60;
        long curretMinutes = totalMinutes % 60;
        long totalHours = totalMinutes / 60;
        long currentHours = totalHours % 60;
        System.out.println("The current time is: " + currentHours +
                ":" + curretMinutes + ":" + currentSeconds);
    }
}
```

**2.14.1** Show the output of the following code:

```
double a = 6.5;
a += a + 1;
System.out.println(a);
```

**Answer:**

a. the output for above code is 14.0 .

```
a = 6;
a /= 2;
System.out.println(a);
```

b. the output for given code is : 3.25 .

**2.15.1** Which of these statements are true?

a. Any expression can be used as a statement.

b. The expression `x++` can be used as a statement.

c. The statement `x = x + 5` is also an expression.

d. The statement `x = y = x = 0` is illegal.

**Answer:** in all above statements (a) is not correct, because in java as well as other programing languages only specific expressions can be turned into statements.
(b) and (c) are correct statements due to java programing language and we can use them.
(d) is also incorrect because it's a completely legal statement that is allowed in Java.

**2.15.2** Show the output of the following code:

```
int a = 6;
int b = a++;
System.out.println(a);
System.out.println(b);
a = 6;
b = ++a;
System.out.println(a);
System.out.println(b);
```

**Answer:** the output of above code is ( 7 , 6 , 7 , 7 ) . for the second code where the output is 6 is because it's a postfix increment and java use its original value in statement.

**2.16.1** Can different types of numeric values be used together in a computation?

**Answer:** yes we can use different types of numeric values like; doubles, integers and etc…. together in a computation.

**2.16.2** What does an explicit casting from a `double` to an `int` do with the fractional part of the `double` value? Does casting change the variable being cast?

**Answer:**

a. when we change the type of a double to an integer the fractional part will be removed because the int data type just store whole numbers.

b. No. casting do not change the variable being cast, it just store a new data which is in another data type

**2.16.3** Show the following output:

```
float f = 12.5F;
int i = (int)f;
System.out.println("f is " + f);
System.out.println("i is " + i);
```

**Answer:** the output will be:

```
f is 12.5
i is 12
```

**2.16.4** If you change `(int)(tax * 100) / 100.0` to `(int)(tax * 100) / 100` in line 11 in Listing 2.8, what will be the output for the input purchase amount of `197.556`?

**Answer:** if we change 100 to 100 it means that we change its data type from double to integer which won't compute its fractional part and output will be:

```
Enter purchase amount: 197.556
Sales tax is $11
```

**2.16.5** Show the output of the following code:

```
double amount = 5;
System.out.println(amount / 2);
System.out.println(5 / 2);
```

**Answer:** in first output 5 is counted as a double so it will include the fractional part as well so the output will be 2.5 , but in second output 5 is evaluated as an integer so it won't contain the fractional part and output will be 2.

```
2.5
2
```

**2.16.6** Write an expression that rounds up a `double` value in variable `d` to an integer.

**Answer:** we can round double values containing fractional parts by changing its type to an integer however it always round down values, so to round up we should use Math.ceil method; but for a simple example we write it as:

```
public class CheckPoints {
    public static void main(String[] args){
        double d = 55.67;
        System.out.println((int) (d + 0.5));
    }
}
```

**2.17.1** How would you write the following arithmetic expression?

$$\frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

**Answer:**

```
public class CheckPoints {
    public static void main(String[] args){
       double a=1, b=5, c=6;
        System.out.print((-b + Math.sqrt(b*b - 4 * a * c))/2*a);
    }
}
```

**2.18.1** Show the output of Listing 2.10 with the input value `1.99`. Why does the program produce an incorrect result for the input 10.03?

**Answer:**

a. The output for 1.99 is:

```
Enter an amount in double, for example 11.56: 1.99
Your amount 1.99 consists of
 1 dollars
 3 quarters
 2 dimes
 0 nickels
 4 pennies
```

b. Here the result for pennies should be 3 but here is 2 which is because of accuracy of double 10.03 so while storing it in computer it may not be stored as 10.03 because if change it to Binary number it will stored a value like: 10.0299 or something near it and it causes 2penies instead of 3penies.

```
Enter an amount in double, for example 11.56: 10.03
Your amount 10.03 consists of
 10 dollars
 0 quarters
 0 dimes
 0 nickels
 2 pennies
```

**2.19.1** Can you declare a variable as `int` and later redeclare it as `double`?

**Answer:** yes we can redeclare a variable so this process is called casting variables, for example: ----------→
In this example we converted its type from int to double.

```
int x = 4;
System.out.print((double) x);
```

**2.19.2** What is an integer overflow? Can floating-point operations cause overflow?

**Answer:**

a. integer overflow is a common error and it happens when we enter a value which is larger than range of data tye we declared variable by.
b. Yes, floating-point operations also will cuse overflow for example: if we enter a value which has more than 17 digits in fractional part it will cause overflow.

**2.19.3** Will overflow cause a runtime error?

**Answer:** yes, because java doesn't report warnings about overflows so when we run our code it will cause error.

**2.19.4** What is a round-off error? Can integer operations cause round-off errors? Can floating-point operations cause round-off errors?

**Answer:**

a. round-off errors which is also known as rounding errors is the difference between the exact value of a number and what computer calculate or store.
b. Yes, the integer operations which its output have a long digits in fractional part will cause this error.
c. Yes, floating points operations also can cause round-off errors and the reason is set in its changing from dicimal to binary.

Chapter 2 checkpoints finished

# Chapter 2 end of unit quiz

**2.1** (*Convert Celsius to Fahrenheit*) Write a program that reads a Celsius degree in a `double` value from the console, then converts it to Fahrenheit, and displays the result. The formula for the conversion is as follows:

```
fahrenheit = (9 / 5) * celsius + 32
```

Hint: In Java, `9 / 5` is `1`, but `9.0 / 5` is `1.8`.

Here is a sample run:

**Answer:**

```
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a value in Celsius: ");
        double value = input.nextDouble();
        double fahrenheit = value *(9.0/5)+32;
        System.out.print(value + " Celsiuses is equal to "
                + fahrenheit + " Fahrenheits");
    }
}
```

**2.2** (*Compute the volume of a cylinder*) Write a program that reads in the radius and length of a cylinder and computes the area and volume using the following formulas:

```
area = radius * radius * π
volume = area * length
```

**Answer:**

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the cylinder's radius: ");
        double radius = input.nextDouble();
        System.out.print("Enter the cylinder's length: ");
        double length = input.nextDouble();
        double area = radius * radius * Math.PI;
        double volume = area * length;
        System.out.println("The area of cylinder's base is: " + area );
        System.out.print("The volume of the cylinder is: " + volume);

    }
}
```

**2.3** (*Convert feet into meters*) Write a program that reads a number in feet, converts it to meters, and displays the result. One foot is `0.305` meter. Here is a sample run:

**Answer:**

```java
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an amount in feet to convert it to meters: ");
        double feet = input.nextDouble();
        double meters = feet * 0.305;
        System.out.print(feet + "feet is equal to " + meters + " meters");
    }
}
```

and the result is:---------->

```
Enter an amount in feet to convert it to meters: 50
50.0feets is equal to 15.25 meters
```

**2.4** (*Convert pounds into kilograms*) Write a program that converts pounds into kilograms. The program prompts the user to enter a number in pounds, converts it to kilograms, and displays the result. One pound is 0.454 kilogram. Here is a sample run:

**Answer**:

```java
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an amount in pounds to convert it into kilograms: ");
        double pounds = input.nextDouble();
        double kilograms = pounds * 0.454;
        System.out.print(pounds + " pounds is equal to " + kilograms + " kilograms");
    }
}
```

and the result is:

```
Enter an amount in pounds to convert it into kilograms: 10
10.0pounds is equal to 4.54 kilograms
Process finished with exit code 0
```

***2.5** (*Financial application: calculate tips*) Write a program that reads the subtotal and the gratuity rate, then computes the gratuity and total. For example, if the user enters 10 for subtotal and 15% for gratuity rate, the program displays $1.5 as gratuity and $11.5 as total. Here is a sample run:

**Answer**:

```java
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the subtotal amount: ");
        double subtotal = input.nextDouble();
        System.out.print("Enter teh gratuity rate in percentage: ");
        double gratuityRate = input.nextDouble();
        double total = subtotal * (gratuityRate/100) + subtotal;
        System.out.print(subtotal + " dollars with gratuity rate of "
                + gratuityRate + " becomes: " + total);
    }
}
```

the output will be:------->

```
Enter the subtotal amount: 10
Enter teh gratuity rate in percentage: 15
10.0 dollars whith gratuity rate of 15.0 becomes: 11.5
```

**2.6 (*Sum the digits in an integer*) Write a program that reads an integer between 0 and 1000 and adds all the digits in the integer. For example, if an integer is 932, the sum of all its digits is 14.

*Hint*: Use the % operator to extract digits, and use the / operator to remove the extracted digit. For instance, 932 % 10 = 2 and 932 / 10 = 93.

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a whole number (0-999) : ");
        int number = input.nextInt();
        int n1 = number % 10;
        int remainder = number / 10;
        int n2 = remainder % 10;
        int n3 = remainder / 10;
        System.out.print("Sum of all digits of " + number + " is: "
           +( n1 + n2 + n3));
    }
}
```

*2.7 (*Find the number of years*) Write a program that prompts the user to enter the minutes (e.g., 1 billion), and displays the maximum number of years and remaining days for the minutes. For simplicity, assume that a year has 365 days. Here is

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the amount o minutes : ");
        double minutes = input.nextDouble();
        double aYearInMinutes = 365 * 24 * 60;
        double year = minutes / aYearInMinutes;
        double days = (minutes % aYearInMinutes) / (24 * 60);
        System.out.print((int)minutes + " minutes is equal to " +
                (int)year + " years and " + (int)days + " days" );
    }
}
```

The output is: ------→

```
Enter the amount o minutes : 1000000000
1000000000 minutes is equal to 1902 years and 214 days
```

*2.8 (*Current time*) Listing 2.7, ShowCurrentTime.java, gives a program that displays the current time in GMT. Revise the program so it prompts the user to enter the time zone offset to GMT and displays the time in the specified time zone. Here is

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter the Enter the time zone in GMT : ");
        float timeZone = input.nextFloat();
        float timeZoneInMilliSeconds = (timeZone * 3600 * 1000);
        long milliSeconds = System.currentTimeMillis();
        long totalMilliseconds = (long)timeZoneInMilliSeconds + milliSeconds;
        long totalSeconds = totalMilliseconds / 1000;
        long currentSecond = totalSeconds % 60;
        long totalMinutes = totalSeconds / 60;
        long currentMinute = totalMinutes % 60;
        long totalHours = totalMinutes / 60;
        long currentHour = totalHours % 24;

        System.out.print("Current time is " + currentHour + ":"
                + currentMinute + ":" + currentSecond + " GMT");
    }
}
```

and now current time in Kabul is:-------→

```
Enter the Enter the time zone in GMT : 4.5
Current time is 10:18:10 GMT
Process finished with exit code 0
```

**2.9** (*Physics: acceleration*) Average acceleration is defined as the change of velocity divided by the time taken to make the change, as given by the following formula:

$$a = \frac{v_1 - v_0}{t}$$

Write a program that prompts the user to enter the starting velocity $v_0$ in meters/second, the ending velocity $v_1$ in meters/second, and the time span $t$ in seconds, then displays the average acceleration. Here is a sample run:

## Answer:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the starting velocity (v0) in (meters/second) : ");
        double startingVelocity = input.nextDouble();
        System.out.print("Enter the ending velocity (v) in (meters/second) : ");
        double endingVelocity = input.nextDouble();
        System.out.print("Enter the time spent: ");
        double time = input.nextDouble();
        System.out.print("The average velocity is: " +
                (endingVelocity - startingVelocity)/time + " meters/seconds");
    }
}
```

the out put is:----------→

```
Enter the starting velocity (v0) in (meters/second) : 50
Enter the ending velocity (v) in (meters/second) : 20
Enter the time spent: 5
The average velocity is: -6.0 meters/seconds
```

**2.10** (*Science: calculating energy*) Write a program that calculates the energy needed to heat water from an initial temperature to a final temperature. Your program should prompt the user to enter the amount of water in kilograms and the initial and final temperatures of the water. The formula to compute the energy is

```
Q = M * (finalTemperature – initialTemperature) * 4184
```

where **M** is the weight of water in kilograms, initial and final temperatures are in degrees Celsius, and energy **Q** is measured in joules. Here is a sample run:

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the weight of water in kilograms : ");
        double M = input.nextDouble();
        System.out.print("Enter the initial temperature of water in Celsius : ");
        double initialTemperature = input.nextDouble();
        System.out.print("Enter the final temperature of water in Celsius : ");
        double finalTemperature = input.nextDouble();
        double Q = M * (finalTemperature - initialTemperature) * 4284;
        System.out.print("The energy needed to heat this amount of water is: " + Q +
" Joules");
    }
}
```

The output is: --→

```
Enter the weight of water in kilograms : 100
Enter the initial temperature of water in Celsius : 12
Enter the final temperature of water in Celsius : 32
The energy needed to heat this amount of water 8568000.0 Joules
```

**2.11** (*Population projection*) Rewrite Programming Exercise 1.11 to prompt the user to enter the number of years and display the population after the number of years. Use the hint in Programming Exercise 1.11 for this program. Here is a sample

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the number of years: ");
        double years = input.nextDouble();
        double population = 312032486;
        double year = 365 * 24 * 3600;
        double birth = year/7;
        double death = year/13;
        double immigrant = year/45;
        double firstYear = population + birth + immigrant - death;
        double secondYear = firstYear + birth + immigrant - death;
        double thirdYear = secondYear + birth + immigrant - death;
        double forthYear = thirdYear + birth + immigrant - death;
        double fifthYear = forthYear + birth + immigrant - death;
        if (years == 1){
            System.out.println("The population after one year will be = " +(int)
firstYear);
        } else if (years == 2) {
            System.out.println("The population after tow years will be = " +(int)
secondYear);
```

```
        } else if (years == 3) {
            System.out.println("The population after three years will be = " +(int)
thirdYear);
        } else if (years == 4) {
            System.out.println("The population after four years will be = " +(int)
forthYear);
        } else if (years == 5) {
            System.out.println("The population after five years will be = " +(int)
fifthYear);
        }
    }
}
```

The output is:---------->

```
Enter the number of years: 3
The population after three years will be = 320372776
```

**2.12** (*Physics: finding runway length*) Given an airplane's acceleration $a$ and take-off speed $v$, you can compute the minimum runway length needed for an airplane to take off using the following formula:

$$\text{length} = \frac{v^2}{2a}$$

Write a program that prompts the user to enter $v$ in meters/second (m/s) and the acceleration $a$ in meters/second squared ($m/s^2$), then, displays the minimum runway length.

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the speed of airplane in m/s: ");
        double speed = input.nextDouble();
        System.out.print("Enter the acceleration of airplane m/s2: ");
        double acceleration = input.nextDouble();
        double runway = (Math.pow(speed, 2))/(2 * acceleration);
        System.out.print("The minimum runway length for this airplane is: " +
runway);
    }
}
```

and the output is: ------>

```
Enter the speed of airplane in m/s: 60
Enter the acceleration of airplane m/s2: 3.5
The minimum runway length for this airplane is: 514.2857142857143
```

**2.13 (*Financial application: compound value*) Suppose you save **$100** *each* month into a savings account with an annual interest rate of 5%. Thus, the monthly interest rate is 0.05/12 = 0.00417. After the first month, the value in the account becomes

```
100 * (1 + 0.00417) = 100.417
```

After the second month, the value in the account becomes

```
(100 + 100.417) * (1 + 0.00417) = 201.252
```

After the third month, the value in the account becomes

```
(100 + 201.252) * (1 + 0.00417) = 302.507
```

and so on.

Write a program that prompts the user to enter a monthly saving amount and displays the account value after the sixth month. (In Programming Exercise 5.30, you will use a loop to simplify the code and display the account value for any month.)

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the amount of money you save monthly: ");
        double amountOfMoney = input.nextDouble();
        final double INEREST_RATE = 0.05/12;
        double firstMonth = amountOfMoney * INEREST_RATE + amountOfMoney;
        double secondMonth = (firstMonth + amountOfMoney) * INEREST_RATE +
(firstMonth + amountOfMoney);
        double thirdMonth = (secondMonth + amountOfMoney) * INEREST_RATE +
(secondMonth + amountOfMoney);
        double fourthMonth = (thirdMonth + amountOfMoney) * INEREST_RATE +
(thirdMonth + amountOfMoney) ;
        double fifthMonth = (fourthMonth + amountOfMoney) * INEREST_RATE +
(fourthMonth + amountOfMoney);
        double sixthMonth = (fifthMonth + amountOfMoney) * INEREST_RATE + (fifthMonth
+ amountOfMoney);
        System.out.print("After the sixth month, the account value is $" +
sixthMonth);
    }
```

The answer is: ----→

```
Enter the amount of money you save monthly: 100
After the sixth month, the account value is $608.8110177055961
```

**2.14 (*Health application: computing BMI*) Body Mass Index (BMI) is a measure of health on weight. It can be calculated by taking your weight in kilograms and dividing, by the square of your height in meters. Write a program that prompts the user to enter a weight in pounds and height in inches and displays the BMI. Note one pound is **0.45359237** kilograms and one inch is **0.0254** meters. Here is a sample run:

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the weight in pounds: ");
        double weight = input.nextDouble();
        System.out.print("Enter the height in inches: ");
        double height = input.nextDouble();
        double heightInMeters = height * 0.0254;
        double weightInKilograms = weight * 0.45359237;
```

```
        double bmi = weightInKilograms/Math.pow(heightInMeters, 2);
        System.out.printf("BMI is: %.2f" , bmi);
    }
}
```

The output is: - ----------→

```
Enter the weight in pounds: 95.5
Enter the height in inches: 50
BMI is: 26.86
```

**2.15** (*Geometry: distance of two points*) Write a program that prompts the user to enter two points `(x1, y1)` and `(x2, y2)` and displays their distance. The formula for computing the distance is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Note you can use `Math.pow(a, 0.5)` to compute $\sqrt{a}$. Here is a sample run:

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter X1 and Y1 : ");
        double x1 = input.nextDouble();  double y1 = input.nextDouble();
        System.out.print("Enter X2 and y2 : ");
        double x2 = input.nextDouble();  double y2 = input.nextDouble();
        double distance = Math.pow(Math.pow((x2 - x1), 2) + Math.pow((y2 - y1),
2),0.5);
        System.out.print("The distance between P(" + x1 + "," + y1 + ") and Q(" + x2
+
                "," + y2 + ") is equal to: " + distance);
    }
}
```

The output is:→

```
Enter X1 and Y1 : 3  4
Enter X2 and y2 : 0 2
The distance between P(3.0,4.0) and Q(0.0,2.0) is equal to: 3.605551275463989
```

**2.16** (*Geometry: area of a hexagon*) Write a program that prompts the user to enter the side of a hexagon and displays its area. The formula for computing the area of a hexagon is

$$\text{Area} = \frac{3\sqrt{3}}{2}s^2,$$

where $s$ is the length of a side. Here is a sample run:

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the length of side of hexagon : ");
        double side = input.nextDouble();
        double area = (3 * (Math.pow(3, 0.5))/2) * Math.pow(side, 2);
        System.out.print("The area of hexagon is equal to: " + area);
    }
}
```

The output is:---------->

```
Enter the length of side of hexagon : 7.7
The area of hexagon is equal to: 154.03993857113812
```

***2.17** (*Science: wind-chill temperature*) How cold is it outside? The temperature alone is not enough to provide the answer. Other factors including wind speed, relative humidity, and sunshine play important roles in determining coldness outside. In 2001, the National Weather Service (NWS) implemented the new wind-chill temperature to measure the coldness using temperature and wind speed. The formula is

$$t_{wc} = 35.74 + 0.6215t_a - 35.75v^{0.16} + 0.4275t_a v^{0.16}$$

where $t_a$ is the outside temperature measured in degrees Fahrenheit, $v$ is the speed measured in miles per hour, and $t_{wc}$ is the wind-chill temperature. The formula cannot be used for wind speeds below 2 mph or temperatures below −58°F or above 41°F.

Write a program that prompts the user to enter a temperature between −58°F and 41°F and a wind speed greater than or equal to **2** then displays the wind-chill temperature. Use `Math.pow(a, b)` to compute $v^{0.16}$. Here is a sample run:

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the temperature in Fahrenheit (-58 to 41) : ");
        double temperature = input.nextDouble();
        System.out.print("Enter the wind speed in mile per hour (>=2) : ");
        double speed = input.nextDouble();
        double coldness = 35.74 + 0.6215 * temperature - 35.75 * Math.pow(speed,
0.16)
                + 0.4275 * temperature * Math.pow(speed, 0.16);
        System.out.print("The wind chill index is: " + coldness);
    }
}
```

The output is: ---------->

```
Enter the temperature in Fahrenheit (-58 to 41) : 7.5
Enter the wind speed in mile per hour (>=2) : 5
The wind chill index is: -1.7007566698129004
```

**2.18** (*Print a table*) Write a program that displays the following table. Cast floating-point numbers into integers.

| a | b | pow(a, b) |
|---|---|---|
| 1 | 2 | 1 |
| 2 | 3 | 8 |
| 3 | 4 | 81 |
| 4 | 5 | 1024 |
| 5 | 6 | 15625 |

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.println("a    b    pow(a,b) ");
        int a, b;
        a=1; b=2;
        System.out.println(a + "    " +  b + "    "  + (int)Math.pow(a, b));
```

```
        a=2; b=3;
        System.out.println(a + "    " + b + "    "  + (int)Math.pow(a, b));
        a=3; b=4;
        System.out.println(a + "    " +  b + "    "  + (int)Math.pow(a, b));
        a=4; b=5;
        System.out.println(a + "    " +  b + "    "  + (int)Math.pow(a, b));
        a=5; b=6;
        System.out.print(a + "    " +  b + "    "  + (int)Math.pow(a, b));
    }
}
```

The output is: ----->

```
a    b    pow(a,b)
1    2    1
2    3    8
3    4    81
4    5    1024
5    6    15625
```

***2.19** (*Geometry: area of a triangle*) Write a program that prompts the user to enter three points, **(x1, y1)**, **(x2, y2)**, and **(x3, y3)**, of a triangle then displays its area. The formula for computing the area of a triangle is

$$s = (\text{side1} + \text{side2} + \text{side3})/2;$$

$$\text{area} = \sqrt{s(s - \text{side1})(s - \text{side2})(s - \text{side3})}$$

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter teh coordinates of three points separated by spaces " +
                "like: x1 y1  x2 y2  x3 y3 : ");
        double x1 = input.nextDouble(); double y1 = input.nextDouble(); double x2 = input.nextDouble();
        double y2 = input.nextDouble(); double x3 = input.nextDouble(); double y3 = input.nextDouble();
        double side1 = Math.pow(Math.pow((x2 - x1), 2) + Math.pow((y2 - y1), 2), 0.5);
        double side2 = Math.pow(Math.pow((x3 - x1), 2) + Math.pow((y3 - y1), 2), 0.5);
        double side3 = Math.pow(Math.pow((x2 - x3), 2) + Math.pow((y2 - y3), 2), 0.5);
        double circumference = side1 + side2 + side3;
        double p = circumference/2;
        double area = Math.pow(p * (p - side1) * (p - side2) * (p-side3), 0.5);
        System.out.print("The area is: " + area);
    }
}
```

The output is: ---->

```
Enter teh coordinates of three points separated by spaces like: x1 y1  x2 y2  x3 y3 : 1.5 -3.4 4.6 5 9.5 -3.4
The area is: 33.600000000000016
```

***2.20** (*Financial application: calculate interest*) If you know the balance and the annual percentage interest rate, you can compute the interest on the next monthly payment using the following formula:

$$\text{interest} = \text{balance} \times (\text{annualInterestRate}/1200)$$

Write a program that reads the balance and the annual percentage interest rate and displays the interest for the next month. Here is a sample run:

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the balance and interest rate (1000  3) : ");
        double balance = input.nextDouble();  double annualInterestRate =
input.nextDouble();
        double interest
= balance *
```

```
Enter the balance and interest rate (1000  3) : 400000  5
The interest is : 1666.6666666666667
```

```
(annualInterestRate/1200);
        System.out.print("The interest is : " + interest);
    }
}
```

The output is:--------→

***2.21** (*Financial application: calculate future investment value*) Write a program that reads in investment amount, annual interest rate, and number of years and displays the future investment value using the following formula:

$$\text{futureInvestmentValue} = \text{investmentAmount} \times (1 + \text{monthlyInterestRate})^{\text{numberOfYears}*12}$$

For example, if you enter amount **1000**, annual interest rate **3.25%**, and number of years **1**, the future investment value is **1032.98**.

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the investment amount: ");
        double investmentAmount = input.nextDouble();
        System.out.print("Enter the annual interest rate: ");
        double annualInterestRate = input.nextDouble();
        double monthlyInterestRate = annualInterestRate/12;
        System.out.print("Enter the number of years: ");
        double numberOfYears = input.nextDouble();
        double futureInvestmentValue = investmentAmount *
           ((Math.pow((1 + monthlyInterestRate/100), (numberOfYears * 12))));
        System.out.print("The future investment is : " + futureInvestmentValue);
    }
}
```

The output is:----------→

```
Enter the investment amount: 100000
Enter the annual interest rate: 10
Enter the number of years: 5
The future investment is : 164530.89347785854
```

***2.22** (*Financial application: monetary units*) Rewrite Listing 2.10, ComputeChange.java, to fix the possible loss of accuracy when converting a **double** value to an **int** value. Enter the input as an integer whose last two digits represent the cents. For example, the input **1156** represents **11** dollars and **56** cents.

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an amount in integer, in which last tow " +
                "digits represent the cents for example 1223 (12 dollars and 23
cents: ");
        double amount1 = input.nextDouble();
        double amount = amount1/100;
        int remainingAmount = (int)(amount * 100);
        int numberOfOneDollars = remainingAmount / 100;
```

```
        remainingAmount = remainingAmount % 100;
        int numberOfQuarters = remainingAmount / 25;
        remainingAmount = remainingAmount % 25;
        int numberOfDimes = remainingAmount / 10;
        remainingAmount = remainingAmount % 10;
        int numberOfNickels = remainingAmount / 5;
        remainingAmount = remainingAmount % 5;
        int numberOfPennies = (int)(amount1%100)%25%10%5;
        System.out.println("Your amount " + amount + " is consists of");
        System.out.println(" " + numberOfOneDollars + " dollars");
        System.out.println(" " + numberOfQuarters + " quarters ");
        System.out.println(" " + numberOfDimes + " dimes");
        System.out.println(" " + numberOfNickels + " nickels");
        System.out.print(" " + numberOfPennies + " pennies");
    }
}
```

***2.23** (*Cost of driving*) Write a program that prompts the user to enter the distance to drive, the fuel efficiency of the car in miles per gallon, and the price per gallon then displays the cost of the trip. Here is a sample run:

**Answer**:

```
import java.util.Scanner;
public class CheckPoints {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Enter the driving distance: ");
        double distance = input.nextDouble();
        System.out.print("Enter the the miles per gallon: ");
        double milesPerGallon = input.nextDouble();
        System.out.print("Enter the price per gallon: ");
        double pricePerGallon = input.nextDouble();
        double cost = (distance/milesPerGallon) * pricePerGallon;
        System.out.print("the cost of driving will be: " + cost);
    }
}
```

The output is:----→

```
Enter Enter the driving distance: 900
Enter the the miles per gallon: 24
Enter the price per gallon: 3
the cost of driving will be: 112.5
```